(b) Using (a), find a UMVUE for α.

5.3.18. Let $X_1, \ldots, X_n$ be a random sample from a population with density function:

$$f(x) = \begin{cases} \dfrac{1}{\theta}, & -\dfrac{\theta}{2} \le x \le \dfrac{\theta}{2}, \quad \theta > 0 \\ 0, & \text{otherwise.} \end{cases}$$

Show that $\left(\min_{1 \le i \le n} X_i, \max_{1 \le i \le n} X_i \right)$ is sufficient for θ.

5.3.19. Let $X_1, \ldots, X_n$ be a random sample from a $G(1, \beta)$ distribution.

(a) Show that $U = \sum_{i=1}^{n} X_i$ is a sufficient statistic for β.

(b) The following is a random sample from a $G(1, \beta)$ distribution:

0.3	3.4	0.4	1.8	0.7	1.0	0.1	2.3	3.7	2.0
0.3	3.7	0.1	1.3	1.2	3.3	0.2	1.3	0.6	0.4

Find a sufficient statistic for β.

5.3.20. Show that X_1 is not sufficient for μ, if $X_1, \ldots, X_n$ is a sample from $N(\mu, 1)$.

5.3.21. Let $X_1, \ldots, X_n$ be a random sample from the truncated exponential distribution with pdf:

$$f(x) = \begin{cases} e^{\theta - x}, & x > \theta \\ 0, & \text{otherwise.} \end{cases}$$

Show that $X_{(1)} = \min(X_i)$ is sufficient for θ.

5.3.22. Let $X_1, \ldots, X_n$ be a random sample from a distribution with pdf:

$$f(x) = \begin{cases} \theta x^{\theta - 1}, & 0 < x < 1, \quad \theta > 0 \\ 0, & \text{otherwise.} \end{cases}$$

Show that $U = X_1, \ldots, X_n$ is a sufficient statistic for θ.

5.3.23. Let $X_1, \ldots, X_n$ be a random sample from a Rayleigh distribution with pdf:

$$f(x) = \begin{cases} \dfrac{2x}{\alpha} e^{-x^2/\alpha}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$$

Show that $\sum_{i=1}^{n} X_i^2$ is sufficient for the parameter α.

5.4 A method of finding the confidence interval: pivotal method

In the previous sections, we studied methods for finding point estimators for the population parameters. In general, the estimates will differ from the true parameter values by varying amounts depending on the sample values obtained. In addition, the point estimates do not convey any measure of reliability.

Now, we discuss another type of estimation, called an *interval estimation.* Although point estimators are useful, interval estimators convey more information about the data that are used to obtain the point estimate. The purpose of using an interval estimator is to have some degree of confidence of securing the true parameter. For an interval estimator of a single parameter θ, we will use the random sample to find two quantities L and U such that $L < \theta < U$ with some probability. Because L and U depend on the sample values, they will be random. This interval (L, U) should have two properties: (1) $P(L < \theta < U)$ is high, that is, the true parameter θ is in (L, U) with high probability, and (2) the length of the interval (L, U) should be relatively narrow on average.

In summary, interval estimation goes a step beyond point estimation by providing, in addition to the estimating interval (L, U), a measure of one's confidence in the accuracy of the estimate. Interval estimators are called *confidence intervals* and the limits are called U and L, the *upper* and *lower confidence limits,* respectively. The associated levels of confidence are

determined by specified probabilities. The width of the CI reflects the amount of variability inherent in the point estimate. Thus, our objective is to find a narrow interval with high probability of enclosing the true parameter, θ. We will restrict our attention to single-parameter estimation.

The probability that a CI will contain the true parameter θ is called the *confidence coefficient.* The confidence coefficient gives the fraction of the time that the constructed interval will contain the true parameter, under repeated sampling.

Let L and U be the lower and upper confidence limits for a parameter θ based on a random sample $X_1, \ldots, X_n$. Both L and U are functions of the sample. We can write the interval estimate of θ as:

$$P(L \leq \theta \leq U) = 1 - \alpha,$$

and we read it as we are $(1 - \alpha)100\%$ confident that the true parameter θ is located in the interval (L, U). The number $1 - \alpha$ is the confidence coefficient, and the interval (L, U) is referred to as a $((1 - \alpha)100\%$ CI) for θ. Thus, if we want a 95% CI for, say, population mean μ, then $\alpha = 0.05$. Note that for the discrete random variables, we may not be able to find a lower bound L and an upper bound U such that the probability, $P(L \leq \theta \leq U)$, is exactly $(1 - \alpha)$. In such a case we can choose L and U such that $P(L \leq \theta \leq U) \geq 1 - \alpha$.

How do we find the CI? For this, we use the error structure of the point estimator to obtain this interval. For instance, we know that the sample mean, $\overline{X}$, is a point estimate (MLE or unbiased estimator) of the population mean μ. In this case, we also know that the standard error of $\overline{X}$ is $\sigma/\sqrt{n}$. If the sample came from a normal population, then for a 95% CI for the mean, multiply the standard error by 1.96 and then add and subtract this product from the sample mean. From this we can also observe that, if everything else remains the same, the size of the CI reduces as the sample size increases.

EXAMPLE 5.4.1

As part of a promotion, the management of a large health club wants to estimate average weight loss for its members within the first 3 months after joining the club. They took a random sample of 45 members of this health club and found that they lost an average of 13.8 lb within the first 3 months of membership with a sample standard deviation of 4.2 lb. Find a 95% CI for the true mean. What if a random sample of 200 members of this health club also resulted in the same sample mean and sample standard deviation?

Solution

Here a point estimate of the true mean μ *is the sample mean* $\bar{x} = 13.8$ *lb. Because* $n = 45$ *is large enough, we can use the central limit theorem (CLT) and use approximate normality for the distribution of* $\overline{X}$ *with mean* μ *and the approximate standard error* $\left(4.2/\sqrt{45}\right) = 0.626$. *Thus a 95% CI is* $13.8 \pm (1.96)(0.626)$, *resulting in the interval* (12.57, 15.03). *Thus, on average, with* 95% *confidence, one can expect the true mean to lie in this interval.*

For $n = 200$, *the standard error is* $\left(4.2/\sqrt{200}\right) \approx 0.297$. *Thus a* 95% *CI is* $13.8 \pm (1.96)(0.297)$ *resulting in the interval* (13.22, 14.38). *Thus, the more sample values (that is, the more information) we have, the tighter (smaller width) the interval.*

The previous example was built on our knowledge of the sampling distribution of the sample mean. What if the sampling distribution of the statistic we are interested in is not readily available? More generally, our success in building CIs for an estimate of a parameter depends on identifying a quantity known as the pivot. We now describe this method.

The *pivotal method* is a general method of constructing a CI using a pivotal quantity. This relies on our knowledge of sampling distributions. Here we have to find a pivotal quantity with the following two characteristics:

(i) It is a function of the random sample (a statistic or an estimator $\widehat{\theta}$) and the unknown parameter θ, where θ is the only unknown quantity, and

(ii) It has a probability distribution that does not depend on the parameter θ.

Suppose that $\widehat{\theta} = \widehat{\theta}(X)$ is a point estimate of θ, and let $p\left(\widehat{\theta}, \theta\right)$ be the pivotal quantity. Now, for a given value of α $(0 < \alpha < 1)$, and constants a and b, with $(a < b)$, let

$$P\left(a \leq p\left(\widehat{\theta}, \theta\right) \leq b\right) = 1 - \alpha.$$

Hence, given $\widehat{\theta}$, the inequality is solved for θ to obtain a region of θ values, usually an interval corresponding to the observed $\widehat{\theta}$ value. This will be a desired CI.

From (i) and (ii), it is important to note that the pivotal quantity depends on the parameter, but its distribution is independent of the parameter. Let $X_1, \ldots, X_n$ be a random sample and let $\widehat{\theta}$ be a reasonable point estimate of θ. For instance, $\widehat{\theta}$ could be the maximum likelihood (or some other) estimator of θ. In general, finding a pivotal quantity may not be easy.

However, if $\widehat{\theta}$ is the sample mean $\overline{X}$ or sample variance S^2, we could find a pivotal quantity with known sampling distributions. Suppose $p\left(\widehat{\theta}, \theta\right)$ is a pivotal quantity with known probability distribution that is independent of θ. (Usually, the probability distribution of the pivotal quantity will be standard normal, t, χ^2, or F distribution.) The following are some of the standard pivotal quantities. If the sample $X_1, \ldots, X_n$ is from $N(\mu, \sigma^2)$:

With μ unknown and σ known, let $\overline{X}$ be the sample mean. Then the pivot is $\left(\overline{X} - \mu\right)/(\sigma/\sqrt{n})$, which has an $N(0, 1)$ distribution (see comments after Corollary 4.2.2).

With μ unknown and σ unknown, then the pivot is $\left(\overline{X} - \mu\right)/(S/\sqrt{n})$, which has a t distribution with $(n - 1)$ degrees of freedom (see Theorem 4.2.9). If n is large, using CLT, the distribution of the pivot is approximately $N(0, 1)$.

If σ^2 is unknown, then the pivot is $(n - 1)S^2/\sigma^2$, which has a χ^2 distribution with $(n - 1)$ degrees of freedom (see Theorem 4.2.8).

The following examples illustrate the pivotal method.

EXAMPLE 5.4.2

Suppose we have a random sample $X_1, \ldots, X_n$ from $N(\mu, 1)$. Construct a 95% CI for μ.

Solution

Here the confidence coefficient is 0.95. We know that the MLE of μ *is* $\overline{X}$, *which has an* $N(\mu, 1/n)$ *distribution. Note that this distribution depends on the unknown value of* μ, *and hence,* $\overline{X}$ *cannot be a pivot. However, taking the* z*-transform of* $\overline{X}$ *we obtain the pivotal quantity as:*

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{\overline{X} - \mu}{1/\sqrt{n}},$$

which has an $N(0, 1)$ *distribution that is a function of the sample measurements and does not depend on* μ. *Hence, this* Z *can be taken as a pivot* $p\left(\widehat{\theta}, \theta\right)$. *Now to find* a *and* b *such that* $P\left(a \leq Z\right) = p\left(\widehat{\theta}, \theta\right) \leq b\right) = 0.95$. *One such choice is to find the value of* a *such that* $P(-a \leq Z \leq a) = 0.95$. *From the normal table,*

$$P\left(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}\right) = 0.95,$$

where $z_{\alpha/2}$ *represents the value of* z *with tail area* $\alpha/2$. *This implies* $a = z_{\alpha/2} = 1.96$. Hence,

$$P(-1.96 \leq Z \leq 1.96) = 0.95,$$

or, using the definition of Z *and solving for* μ, *we obtain:*

$$P\left(\overline{X} - \frac{1.96}{\sqrt{n}} \leq \mu \leq \overline{X} + \frac{1.96}{\sqrt{n}}\right) = 0.95.$$

Hence, a 95% CI for μ *is* $\left(\overline{X} - (1.96/\sqrt{n}),\ \overline{X} + (1.96/\sqrt{n})\right)$ *Thus, the lower confidence limit* L *is* $\overline{X} - (1.96/\sqrt{n})$ *and the upper confidence limit* U *is* $\overline{X} + (1.96/\sqrt{n})$.

From the derivation of Example 5.4.1, it follows that:

$$P\left(\left|\overline{X} - \mu\right| < z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

Thus, for a normal population with known variance σ^2, if $\overline{X}$ is used as an estimator of the true mean μ, the probability that the error will be less than $z_{\alpha/2}\sigma/\sqrt{n}$ is $1 - \alpha$. It is important to note that there is some arbitrariness in choosing a CI for a given problem. There may be several pivotals for $\widehat{\theta}$ that could be used. Also, it is not necessary to allocate equal probability to the two tails of the distribution; however, doing so may result in the shortest length CI for a given confidence coefficient.

When we make the statement of the form:

$$P\left(\overline{X} - \frac{1.96}{\sqrt{n}} \leq \mu \leq \overline{X} + \frac{1.96}{\sqrt{n}}\right) = 0.95,$$

we mean that, in an infinite series of trials in which repeated samples of size n are drawn from the same population and 95% CIs for μ are calculated by the same method for each of the samples, the proportion of intervals that actually include μ will be 0.95. Fig. 5.4 illustrates this idea, where the vertical line represents the position of true mean μ and each of the horizontal lines represents a 95% CI of the sample, and 20 samples of size n are taken.

A statement of the type $P(\overline{x}-(1.96/\sqrt{n}) \leq \mu \leq \overline{x}+(1.96/\sqrt{n})) = 0.95$, where $\overline{x}$ is the observed sample mean, is misleading. Once we calculate this interval using a particular sample, then either this interval contains the true mean μ or not, and hence, the probability will be either 0 or 1. Thus, the correct interpretation of CI for the population mean is that if samples of the same size, n, are drawn repeatedly from a population, and a CI is calculated from each sample, then 95% of these intervals should contain the population mean. This is often stated as "We are 95% confident that the true mean is in the interval $\left(\overline{X}-z_{\alpha/2}\left(\sigma/\sqrt{n}\right), \overline{X}-z_{\alpha/2}\left(\sigma/\sqrt{n}\right)\right)$." This concept of CI is attributed to Neyman.

We can follow the accompanying procedure to find a CI for the parameter θ.

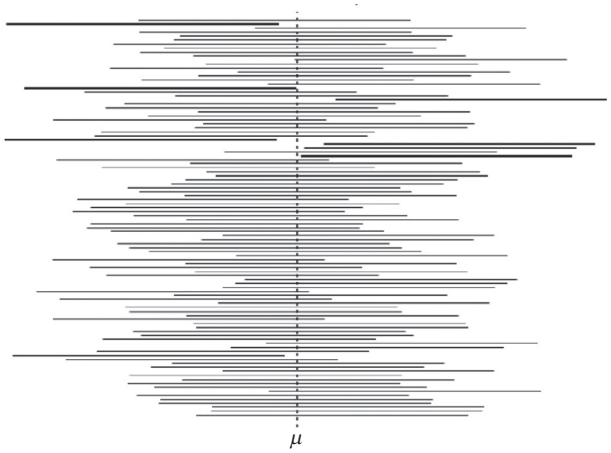


FIGURE 5.4 The 95% confidence intervals for μ.

Procedure to find a confidence interval for θ using the pivot

1. Find an estimator $\widehat{\theta}$ of θ: usually the MLE of θ works.
2. Find a function of θ and $\widehat{\theta}, p\left(\theta, \widehat{\theta}\right)$ (pivot), such that the probability distribution of $p(.,.)$ does not depend on θ.
3. Find a and b such that $P\left(a \leq p\left(\theta, \widehat{\theta}\right) \leq b\right) = 1-\alpha$. Choose a and b such that $P\left(p\left(\theta, \widehat{\theta}\right) \leq a\right) = \alpha/2$ and $P\left(p\left(\theta, \widehat{\theta}\right) \geq b\right) = \alpha/2$ (see Fig. 5.5 where the shaded area in each side is $\alpha/2$).
4. Now, transform the pivot CI to a CI for the parameter θ. That is, work with the inequality in step 3 and rewrite it as $P(L \leq \theta \leq U) = 1-\alpha$, where L is the lower confidence limit and U is the upper confidence limit.

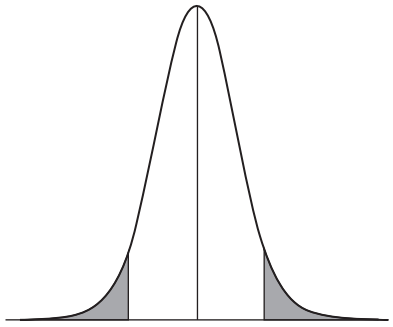


FIGURE 5.5 Probability density of the pivot.

The following example is given to show that the success of finding a pivotal quantity depends on our ability to find the right transformation of the statistic and its distribution so that the transformed variable is a pivot.

EXAMPLE 5.4.3

Suppose the random sample $X_1, \ldots, X_n$ has $U(0, \theta)$ distribution. Construct a 90% CI for θ and interpret. Identify the upper and lower confidence limits.

Solution

From Example 5.3.4, we know that:

$$U = \max_{1 \le i \le n} X_i$$

is the MLE of θ. *The random variable* U *has the pdf:*

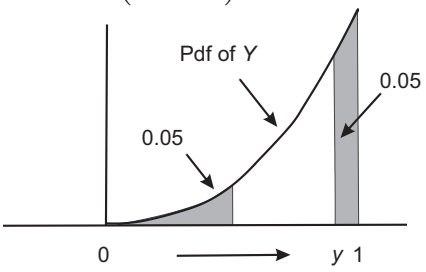
$$f_U(u) = nu^{n-1}/\theta^n, \quad 0 \le u \le \theta.$$

This is not independent of the parameter θ. *Let* $Y = U/\theta$, *then (using the Jacobians described in Chapter 3) the pdf of* Y *is given by:*

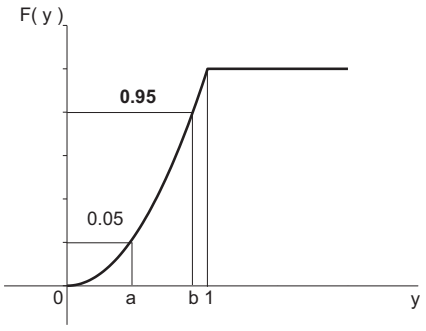
$$f_Y(y) = ny^{n-1}, \quad 0 \le y \le 1.$$

Hence, Y *satisfies the two characteristics of the pivotal quantity. Thus,* $Y = U/\theta$ *is a pivot. Now, we have to find* a *and* b *such that:*

$$p\left(a \le \frac{U}{\theta} \le b\right) = 0.90.$$



To find a *and* b *we use the cdf of* Y, $F_Y(y) = y^n$, $0 \le y \le 1$, *as follows:*



$$F_Y(a) = 0.05 \quad \text{and} \quad F_Y(b) = 0.95,$$

which implies that:

$$a^n = 0.05 \quad \text{and} \quad b^n = 0.95$$

resulting in:

$$a = \sqrt[n]{0.05} \quad \text{and} \quad b = \sqrt[n]{0.95}.$$

Write:

$$P\left(\sqrt[n]{0.05} < \frac{U}{\theta} < \sqrt[n]{0.95}\right) = 0.90.$$

Solving, the 90% CI for θ *is:*

$$\left(\frac{U}{\sqrt[n]{0.95}}, \frac{U}{\sqrt[n]{0.05}}\right)$$

or

$$P\left(\frac{U}{\sqrt[n]{0.95}} \le \theta \le \frac{U}{\sqrt[n]{0.05}}\right) = 0.90.$$

Thus, the lower confidence limit is $U/\sqrt[n]{0.95}$ *and the upper confidence limit is* $U/\sqrt[n]{0.05}$, *and the 90% CI is* $\left(U/\sqrt[n]{0.95},\ U/\sqrt[n]{0.05}\right)$

We can interpret this in the following manner. In a large number of trials in which repeated samples are taken from a population with uniform pdf with parameter θ, approximately 90% of the intervals will contain θ. For instance, if we observed $n = 20$ values from a uniform distribution with the maximum observed value being 15, then a 90% CI for θ is (15.04, 17.42). Thus, we are 90% confident that these data came from a uniform distribution upper limit falling somewhere in this interval.

It is important to note that the pivotal method may not be applicable in all situations. For example, in the binomial case, to find a CI for p, there is no quantity that satisfies the two conditions of a pivot. However, if the sample size is large, then the z-score of sample proportion can be used as a pivot with approximate standard normal distribution. For the pivotal method to work, there is the practical necessity that the distribution of the pivotal quantity make it easy to compute the probabilities. In cases where the pivotal method does not work, we may need to use other techniques such as the method based on sampling distributions (see Project 4A). A proper discussion of these methods is beyond the level of this book.

EXERCISES 5.4

5.4.1. **(a)** Suppose we construct a 99% CI. What are we 99% confident about?
(b) Which of these CIs is wider, 90% or 99%?
(c) In computing a CI, when do you use the t distribution and when do you use z, with normal approximation?
(d) How does the sample size affect the width of a CI?

5.4.2. Suppose X is a random sample of size $n = 1$ from a uniform distribution defined on the interval $(0, \theta)$. Construct a 98% CI for θ and interpret.

5.4.3. Consider the probability statement:

$$P\left(-2.81 \le Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \le 2.75\right) = \kappa,$$

where $\overline{X}$ is the mean of a random sample of size n from an $N(\mu, \sigma^2)$ distribution with known σ^2.
(a) Find κ.
(b) Use this statement to find a CI for μ.
(c) What is the confidence level of this CI?
(d) Find a symmetric CI for μ.

5.4.4. A random sample of size 50 from a particular brand of 16-oz tea packets produced a mean weight of 15.65 oz. Assume that the weights of these brands of tea packets are normally distributed with standard deviation of 0.59 oz. Find a 95% CI for the true mean μ.

5.4.5. Let $X_1, \ldots, X_n$ be a random sample from an $N(\mu, \sigma^2)$, where the value of σ^2 is unknown.
(a) Construct a $(1 - \alpha)100\%$ CI for σ^2, choosing an appropriate pivot. Interpret its meaning.
(b) Suppose a random sample from a normal distribution gives the following summary statistics: $n = 21$, $\overline{x} = 44.3$, and $s = 3.96$. Using (a), find a 90% CI for σ^2. Interpret its meaning.

5.4.6. Let $X_1, \ldots, X_n$ be a random sample from a gamma distribution with $\alpha = 2$ and unknown β. Construct a 95% CI for β.

5.4.7. Let $X_1, \ldots, X_n$ be a random sample from an exponential distribution with pdf $f(x) = (1/\theta)e^{-x/\theta}$, $\theta > 0$, $x > 0$. Construct a 95% CI for θ and interpret. (Hint: Recall that $\sum_{i=1}^{n} X_i$ has a gamma distribution with $\alpha = n$, $\beta = \theta$.)

5.4.8. Let $X_1, \ldots, X_n$ be a random sample from a Poisson distribution with parameter λ.
(a) Construct a 90% CI for λ.

(b) Suppose that the number of raisins in a bowl of a particular brand of cereal is observed to be 25. Assuming that the number of raisins in a bowl is Poisson distributed, estimate the expected number of raisins per bowl with a 90% CI.

(c) How many bowls of cereal need to be sampled to estimate the expected number of raisins per bowl with a standard error of less than 0.2?

5.4.9. Let $X_1, \ldots, X_n$ be a random sample from an $N(\mu, \sigma^2)$.

(a) Construct a $(1-\alpha)100\%$ CI for μ when the value of σ^2 is known.

(b) Construct a $(1-\alpha)100\%$ CI for μ when the value of σ^2 is unknown.

5.4.10. Let $X_1, \ldots, X_n$ be a random sample from an $N(\mu_1, \sigma^2)$ population and $Y_1, \ldots, Y_n$ be an independent random sample from an $N(\mu_2, \sigma^2)$ distribution where σ^2 is assumed to be known. Construct a $(1-\alpha)100\%$ interval for $(\mu_1 - \mu_2)$. Interpret its meaning.

5.4.11. Let $X_1, \ldots, X_n$ be a random sample from a uniform distribution on $[\theta, \theta+1]$. Find a 99% CI for θ, using an appropriate pivot.

5.5 One-sample confidence intervals

In this section, we will find CIs for the one-sample case for both large- and small-sample situations.

5.5.1 Large-sample confidence intervals

If the sample size is large, then by the CLT, certain sampling distributions can be assumed to be approximately normal. That is, if θ is an unknown parameter (such as μ, p, $(\mu_1 - \mu_2)$, $(p_1 - p_2)$), then for large samples, by the CLT, the z-transform:

$$z = \frac{\widehat{\theta} - \theta}{\sigma_{\widehat{\theta}}},$$

possesses an approximately standard normal distribution, where $\widehat{\theta}$ is the MLE of θ and $\sigma_{\widehat{\theta}}$ is its standard deviation. Then as in Example 5.4.1, the pivotal method can be used to obtain the CI for the parameter θ. For $\theta = \mu$, $n \geq 30$ will be considered large; for the binomial parameter p, n is considered large if np and $n(1-p)$ are both greater than 5. Note that these numbers are only a rule of thumb.

Procedure to calculate large-sample confidence interval for θ

1. Find an estimator (such as the MLE) of θ, say $\widehat{\theta}$.
2. Obtain the standard error, $\sigma_{\widehat{\theta}}$ of $\widehat{\theta}$.
3. Find the z-transform $z = \left(\widehat{\theta} - \theta\right) \Big/ \sigma_{\widehat{\theta}}$. Then z has an approximately standard normal distribution.
4. Using the normal table, find two tail values $-z_{\alpha/2}$ and $z_{\alpha/2}$.
5. An approximate $(1-\alpha)100\%$ CI for θ is $\left(\widehat{\theta} - z_{\alpha/2}\sigma_{\widehat{\theta}}, \widehat{\theta} + z_{\alpha/2}\sigma_{\widehat{\theta}}\right)$, that is,

$$P\left(\widehat{\theta} - z_{\alpha/2}\sigma_{\widehat{\theta}} \leq \theta \leq \widehat{\theta} + z_{\alpha/2}\sigma_{\widehat{\theta}}\right) = 1 - \alpha.$$

6. *Conclusion:* We are $(1-\alpha)100\%$ confident that the true parameter θ lies in the interval $\left(\widehat{\theta} - z_{\alpha/2}\sigma_{\widehat{\theta}}, \widehat{\theta} + z_{\alpha/2}\sigma_{\widehat{\theta}}\right)$

EXAMPLE 5.5.1

Let $\widehat{\theta}$ be a statistic that is normally distributed with mean θ and standard deviation $\sigma_{\widehat{\theta}}$, where σ is assumed to be known. Find a CI for θ that possesses a confidence coefficient equal to $1-\alpha$.

Solution

The z-transform of $\widehat{\theta}$ *is:*

$$Z = \frac{\widehat{\theta} - \theta}{\sigma_{\widehat{\theta}}}$$

and has a standard normal distribution. Select two tail values $-z_{\alpha/2}$ *and* $z_{\alpha/2}$ *such that:*

$$P\left(-z_{\alpha/2} \le Z \le z_{\alpha/2}\right) = 1 - \alpha.$$

Because of symmetry, this is the shortest interval that contains the area $1 - \alpha$*. Then,*

$$P\left(\widehat{\theta} - z_{\alpha/2}\sigma_{\widehat{\theta}} \le \theta \le \widehat{\theta} + z_{\alpha/2}\sigma_{\widehat{\theta}}\right) = 1 - \alpha.$$

Therefore, the confidence limits of θ *are* $\widehat{\theta} - z_{\alpha/2}\sigma_{\widehat{\theta}}$ *and* $\widehat{\theta} + z_{\alpha/2}\sigma_{\widehat{\theta}}$*. Hence, the* $(1 - \alpha)100\%$ *CI for* θ *is given by* $\widehat{\theta} \pm z_{\alpha/2}\sigma_{\widehat{\theta}}$.

In particular, for a large sample of size n, let $\widehat{\theta} = \overline{X}$ be the sample mean. Then the large-sample $(1 - \alpha)100\%$ CI for the population mean μ is:

$$\overline{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \simeq \overline{X} \pm z_{\alpha/2}\frac{S}{\sqrt{n}}$$

where S is a point estimate of σ. That is,

$$P\left(\overline{X} - z_{\alpha/2}\frac{S}{\sqrt{n}} \le \mu \le \overline{X} \pm z_{\alpha/2}\frac{S}{\sqrt{n}}\right) = 1 - \alpha.$$

As we have seen in Section 5.4, the correct interpretation of this CI is that in a repeated sampling, approximately $(1 - \alpha)100\%$ of all intervals of the form $\overline{X} \pm z_{\alpha/2}(S/\sqrt{n})$ include μ, the true mean. Suppose $\bar{x}$ and s are the sample mean and the sample standard deviation, respectively, for a particular set of n observed sample values $x_1, \ldots, x_n$. Then we do not know whether the particular interval $\left(\bar{x} - z_{\alpha/2}(s/\sqrt{n}),\ \bar{x} - z_{\alpha/2}(s/\sqrt{n})\right)$ contains μ. However, the procedure that produced this interval does capture the true mean in approximately $(1 - \alpha)100\%$ of cases. This interpretation will be assumed hereafter, when we make a statement such as, "We are 95% confident that the true mean will lie in the interval (74.1, 79.8)."

EXAMPLE 5.5.2

Two statistics professors want to estimate average scores for an elementary statistics course that has two sections. Each professor teaches one section and each section has a large number of students. A random sample of 50 scores from each section produced the following results:

(a) Section I: $\bar{x}_1 = 77.01$, $s_1 = 10.32$

(b) Section II: $\bar{x}_2 = 72.22$, $s_2 = 11.02$

Calculate 95% CIs for each of these two samples.

Solution

Because $n = 50$ *is large, we could use normal approximation. For* $\alpha = 0.05$*, from the normal table:* $z_{\alpha/2} = z_{0.025} = 1.96$*. The CIs are as follows:*

(a) *We have:*

$$\bar{x}_1 \pm z_{\alpha/2}\frac{s_1}{\sqrt{n}} = 77.01 \pm 1.96\left(\frac{10.32}{\sqrt{50}}\right),$$

which gives a 95% CI (74.149, 79.871).

(b) *We can compute:*

$$\bar{x}_2 \pm z_{\alpha/2}\frac{s_2}{\sqrt{n}} = 72.22 \pm 1.96\left(\frac{11.02}{\sqrt{50}}\right),$$

which gives the interval (69.165, 75.275).

It may be noted that if the population is normal with a known variance σ^2, we can use $\overline{X} \pm z_{\alpha/2}(\sigma/\sqrt{n})$ as the CI for the population mean μ, irrespective of the sample size. However, if σ^2 is unknown, to use $\overline{X} \pm z_{\alpha/2}(s/\sqrt{n})$ as an approximate CI for μ, the sample size has to be large for the CLT to hold. However, to use this approximate procedure, we do not need the condition that samples arise from a normal distribution. We will consider sample size to be large if $n \ge 30$ (applicable to estimators of the mean). If not, we shall use the small-sample procedure discussed in the next section.

EXAMPLE 5.5.3

Fifteen vehicles were observed at random for their speeds (in mph) on a highway with speed limit posted as 70 mph, and it was found that their average speed was 73.3 mph. Suppose that from past experience we can assume that vehicle speeds are normally distributed with $\sigma = 3.2$. Construct a 90% CI for the true mean speed μ, of the vehicles on this highway. Interpret the result.

Solution

Because the population is given to be normal with standard deviation $\sigma = 3.2$, *sample size need not be large, given* $\bar{x} = 73.3$ *and* $\sigma = 3.2$. *Here,* $n = 15$, *and* $\alpha = 0.10$. *Thus,* $z_{\alpha/2} = z_{0.05} = 1.645$. *Hence, a 90% CI for* μ *is given by:*

$$73.3 - 1.645\frac{3.2}{\sqrt{15}} < \mu < 73.3 + 1.645\frac{3.2}{\sqrt{15}}$$

or

$$71.681 < \mu < 74.919.$$

Interpretation: We are 90% confident that the true mean speed μ *of the vehicles on this highway is between 71.681 and 74.919 mph.*

5.5.2 Confidence interval for proportion, *p*

Consider a binomial distribution with parameter p. Let X be the number of successes in n trials. Then the MLE $\widehat{p}$ of p is $\widehat{p} = X/n$. It can be shown, using the procedure outlined at the beginning of this section, that an approximate large-sample $(1 - \alpha)100\%$ CI for p is:

$$\left(\widehat{p} - z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}, \widehat{p} + z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}\right).$$

That is,

$$P\left(\widehat{p} - z_{\alpha/2}\sqrt{\frac{\widehat{p}1-\widehat{p})}{n}} < p < \widehat{p} + z_{\alpha/2}\sqrt{\frac{\widehat{p}1-\widehat{p})}{n}}\right) = 1 - \alpha.$$

A natural question is, "How do we determine that the sample size we have is sufficient for the normal approximation that is used in the foregoing formula?" There are various rules of thumb that are used to determine the adequacy of the sample size for normal approximation. Some of the popular rules are that np and $n(1 - p)$ should be greater than 10, or that $\widehat{p} \pm 2\sqrt{\widehat{p}(1-\widehat{p})/n}$ should be contained in the interval (0,1), or $np(1 - p) \geq 10$, etc. All of these rules perform poorly when p is nearer to 0 or 1. There have been many works on coverage analysis for CIs. We refer to a survey article by Lee et al., for more details on this topic. For simplicity of calculations, we will use the rule that np and $n(1 - p)$ are both greater than 5.

EXAMPLE 5.5.4

An auto manufacturer gives a bumper-to-bumper warranty for 3 years or 36,000 miles for its new vehicles. In a random sample of 60 of its vehicles, 20 of them needed five or more major warranty repairs within the warranty period. Estimate the true proportion of vehicles from this manufacturer that need five or more major repairs during the warranty period, with confidence coefficient 0.95. Interpret.

Solution

Here we need to find a 95% CI for the true proportion, p. *Here,* $\widehat{p} = 20/60 = 1/3$. *For* $\alpha = 0.05$, $z_{\alpha/2} = z_{0.025} = 1.96$. *Hence, a* 95% *CI for* p *is:*

$$\widehat{p} \pm z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} = \frac{1}{3} \pm 1.96\sqrt{\frac{\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)}{60}},$$

which gives the CI as (0.21405, 0.45262). *That is, we are* 95% *confident that the true proportion of vehicles from this manufacturer that need five or more major repairs during the warranty period will lie in the interval* (0.21405, 0.45262).

5.5.2.1 Margin of error and sample size

In real-world problems, the estimates of the proportion p are usually accompanied by a margin of error, rather than a CI. For example, in the news media, especially leading up to election time, we hear statements such as "The CNN/USA Today/Gallup poll of 818 registered voters taken on June 27−30 showed that if the election were held now, the president would beat his challenger 52% to 40%, with 8% undecided. The poll had a margin of error of plus or minus 4 percentage points." What is this "margin of error"? According to the American Statistical Association, the margin of error is a common summary of sampling error that quantifies uncertainty about a survey result. Thus, the margin of error is nothing but a CI. The number quoted in the foregoing statement is half the maximum width of a 95% CI, expressed as a percentage.

Let b be the width of a 95% CI for the true proportion, p. Let $\widehat{p} = x/n$ be an estimate for p where x is the number of successes in n trials. Then,

$$b = \frac{x}{n} + 1.96\sqrt{\frac{(x/n)(1-(x/n))}{n}} - \left(\frac{x}{n} - 1.96\sqrt{\frac{(x/n)(1-(x/n))}{n}}\right)$$

$$= 3.92\sqrt{\frac{(x/n)(1-(x/n))}{n}} \leq 3.92\sqrt{\frac{1}{4n}},$$

because $(x/n)(1-(x/n)) = \widehat{p}(1-\widehat{p}) \leq \frac{1}{4}$.

Thus, the margin of error associated with $\widehat{p} = (x/n)$ is $100d\%$, where:

$$d = \frac{\max b}{2} = \frac{3.92\sqrt{\frac{1}{4n}}}{2} = \frac{1.96}{2\sqrt{n}}.$$

From the foregoing derivation, it is clear that we can compute the margin of error for other values of α by replacing 1.96 with the corresponding value of $z_{\alpha/2}$.

A quick look at the formula for the CI for proportions reveals that a larger sample would yield a shorter interval (assuming other things being equal) and hence, a more precise estimate of p. The larger sample is costlier in terms of time, resources, and money, whereas samples that are too small may result in inaccurate inferences. Then, it becomes beneficial for finding out the minimum sample size required (thus less costly) to achieve a prescribed degree of precision (usually, the minimum degree of precision acceptable). We have seen that the large-sample $(1-\alpha)100\%$ CI for p is:

$$\widehat{p} - z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} < p < \widehat{p} + z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.$$

Rewriting it, we have:

$$|\widehat{p} - p| \leq z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} = \frac{z_{\alpha/2}}{\sqrt{n}}\sqrt{\widehat{p}(1-\widehat{p})},$$

which shows that, with probability $(1-\alpha)$, the estimate $\widehat{p}$ is within $z_{\alpha/2}\sqrt{\widehat{p}(1-\widehat{p})/n}$ units of p. Because $\widehat{p}(1-\widehat{p}) \leq 1/4$, for all values of $\widehat{p}$, we can write the foregoing inequality as:

$$|\widehat{p} - p| \leq \frac{z_{\alpha/2}}{\sqrt{n}}\sqrt{\frac{1}{4}} = \frac{z_{\alpha/2}}{2\sqrt{n}}.$$

If we wish to estimate p at level $(1-\alpha)$ to within d units of its true value, that is $|\widehat{p} - p| \leq d$, the sample size must satisfy the condition $\left(z_{\alpha/2}/(2\sqrt{n})\right) \leq d$, or

$$n \geq \frac{z_{\alpha/2}^2}{4d^2}.$$

Thus, to estimate p at level $(1-\alpha)$ to within d units of its true value, take the minimal sample size as $n = z_{\alpha/2}^2/(4d^2)$, and if this is not an integer, round up to the next integer.

Sometimes, we may have an initial estimate $\widetilde{p}$ of the parameter p from a similar process or from a pilot study or simulation. In this case, we can use the following formula to compute the minimum required size of the sample to estimate p, at level $(1-\alpha)$, to within d units:

$$n = \frac{z^2_{\alpha/2}\widetilde{p}(1-\widetilde{p})}{d^2}$$

and, if this is not an integer, we round up to the next integer.

A similar derivation for calculation of sample size for estimation of the population mean μ at level $(1-\alpha)$ with margin of error E is given by:

$$n = \frac{z^2_{\alpha/2}\sigma^2}{E^2}$$

and, if this is not an integer, rounding up to the next integer. This formula can be used only if we know the population standard deviation, σ. Although it is unlikely we will know σ when the population mean itself is not known, we may be able to determine σ from an earlier similar study or from a pilot study/simulation.

EXAMPLE 5.5.5

A dendritic tree is a branched formation that originates from a nerve cell. To study brain development, researchers want to examine the brain tissues from adult guinea pigs. How many cells must the researchers select (randomly) so as to be 95% sure that the sample mean is within 3.4 cells of the population mean? Assume that a previous study has shown $\sigma = 10$ cells.

Solution

A 95% *confidence corresponds to* $\alpha = 0.05$. *Thus, from the normal table,* $z_{\alpha/2} = z_{0.025} = 1.96$. *Given that* $E = 3.4$ *and* $\sigma = 10$, *and using the sample size formula, the required sample size* n *is:*

$$n = \frac{z^2_{\alpha/2}\sigma^2}{E^2} = \frac{(1.96)^2(10)^2}{(3.4)^2} = 33.232.$$

Thus, take $n = 34$.

EXAMPLE 5.5.6

Suppose that a local TV station in a city wants to conduct a survey to estimate support for the president's policies on the economy within 3% error with 95% confidence.

(a) How many people should the station survey if they have no information on the support level?

(b) Suppose they have an initial estimate that 70% of the people in the city support the economic policies of the president. How many people should the station survey?

Solution

Here $\alpha = 0.05$, *and thus* $z_{\alpha/2} = 1.96$. *Also,* $d = 0.03$.

(a) *With no information on* p, *we use the sample size formula:*

$$n = \frac{z^2_{\alpha/2}}{4d^2} = \frac{(1.96)^2}{4(0.03)^2} = 1067.1.$$

Hence, the TV station must survey 1068 people.

(b) *Because* $\widetilde{p} = 0.7$, *the required sample size is calculated from:*

$$n = \frac{z^2_{\alpha/2}\widetilde{p}(1-\widetilde{p})}{d^2}$$

$$= \frac{(1.96)^2(0.70)(0.30)}{(0.03)^2} = 896.37.$$

Thus, the TV station must survey at least 897 people.

In practice, we should realize that one of the key factors of a good design is not sample size by itself; it is getting representative samples. Even if we have a very large sample size, if the sample is not representative of our target

population, then sample size means nothing. Therefore, whenever possible, we should use random sampling procedures (or other appropriate sampling procedures) to ensure that our target population is properly represented.

5.5.3 Small-sample confidence intervals for μ

Now we will consider the problem of finding a CI for the true mean μ of a normal population when the variance σ^2 is unknown and obtaining a large sample is either impossible or impractical. Let $X_1, \ldots, X_n$ be a random sample from a normal population. We know that:

$$T = \frac{\sqrt{n}\,\dfrac{\overline{X}-\mu}{\sigma}}{\sqrt{(n-1)S^2/[\sigma^2(n-1)]}} = \frac{\overline{X}-\mu}{S/\sqrt{n}}$$

has a t distribution with $(n-1)$ degrees of freedom, irrespective of the value of σ^2. Thus, $\left(\overline{X}-\mu\right)/\left(S/\sqrt{n}\right)$ can be used as a pivot. Hence, for n small ($n < 30$) and σ^2 unknown, we have the following result.

Theorem 5.5.1 *If* $\overline{\mathrm{X}}$ *and* S *are the sample mean and the sample standard deviation of a random sample of size* n *from a normal population, then:*

$$\overline{X} - t_{\alpha/2.n-1}\frac{S}{\sqrt{n}} < \mu < \overline{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$$

is a $(1-\alpha)100\%$ *CI for the population mean* μ.

Note that if the confidence coefficient, $1-\alpha$, and $\overline{X}$ and S remain the same, the confidence range $CR = \widehat{\theta}_U - \widehat{\theta}_L$ decreases as the sample size n increases, which means that we are closing in on the true parameter value of θ.

One can use the following procedure to find the CI for the mean when a small sample is from an approximately normal distribution.

Procedure to find small-sample confidence interval for μ

1. Calculate the values of $\overline{X}$ and S from the sample $X_1, \ldots, X_n$.
2. Using the t table, select two tail values, $-t_{\alpha/2}$ and $t_{\alpha/2}$.
3. The $(1-\alpha)100\%$ CI for μ is:

$$\left(\overline{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}, \overline{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}\right)$$

that is, $P\left(\overline{X} - t_{\alpha/2,n-1}\dfrac{S}{\sqrt{n}} \le \mu \le \overline{X} + t_{\alpha/2,n-1}\dfrac{S}{\sqrt{n}}\right) = 1-\alpha.$

4. *Conclusion*: We are $(1-\alpha)100\%$ confident that the true parameter μ lies in the interval $\left(\overline{X} - t_{\alpha/2,n-1}(S/\sqrt{n}), \overline{X} + t_{\alpha/2,n-1}(S/\sqrt{n})\right)$.
5. **Assumption:** The population is normal.

In practice, the first step in the previous procedure should include a test of normality (see Project 4C). A built-in test of normality is available in most of the statistical software packages. In Example 5.5.9, we show how this test is utilized. Even when the data fail the normality test, most statistical software will produce a CI based on normality or give an error report. We should understand that generally such answers are meaningless. In those cases, nonparametric methods (Chapter 12) such as the Wilcoxon rank sum method or bootstrap method (Chapter 13) will be more appropriate. For more discussion, refer to Section 14.4.1.

EXAMPLE 5.5.7

The following is a random data set from a normal population:

7.2 5.7 4.9 6.2 8.5 2.8

Construct a 95% CI for the population mean μ. Interpret.

Solution

The first step is to calculate mean and standard deviation of the sample. We compute as the mean $\overline{\mathrm{x}} = 5.883$ *and as standard deviation,* $\mathrm{s} = 1.959$. *For 5 degrees of freedom, and for* $\alpha = 0.05$, *from the* t *table,* $\mathrm{t}_{0.025} = 2.571$. *Hence, a* 95% *CI for* μ *is:*

$$\left(\overline{x} - t_{\alpha/2,n-1}\frac{2}{\sqrt{n}}, \overline{x} + t_{\alpha/2,n-1}\frac{2}{\sqrt{n}}\right)$$

$$= \left(5.883 - 2.571\left(\frac{1.959}{\sqrt{6}}\right), 5.5883 + 2.571\left(\frac{1.959}{\sqrt{6}}\right)\right)$$

$$= (3.827,\ 7.939).$$

This can be interpreted as that we are 95% *confident that the true mean* μ *will be between* 3.827 *and* 7.939.

EXAMPLE 5.5.8

The scores of a random sample of 16 people who took the TOEFL (Test of English as a Foreign Language) had a mean of 540 and a standard deviation of 50. Construct a 95% CI for the population mean μ of the TOEFL score, assuming that the scores are normally distributed.

Solution

Because n = 16 *is small, using Theorem* 5.5.1 *with degrees of freedom* 15*, a* 95% *CI for* μ *is:*

$$\overline{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 540 \pm 2.131\left(\frac{50}{\sqrt{16}}\right).$$

So the 95% *CI for the population mean* μ *of the TOEFL scores is* (513.36, 566.64).

A Dobson unit is the most basic measure used in ozone research. The unit is named after G.M.B. Dobson, one of the first scientists to investigate atmospheric ozone (between 1920 and 1960). He designed the Dobson spectrometer, the standard instrument used to measure ozone from the ground. The data in Example 5.5.9 represent the total ozone levels at randomly selected points on the Earth (represented by the pair (latitude, longitude)) on a particular day.

EXAMPLE 5.5.9

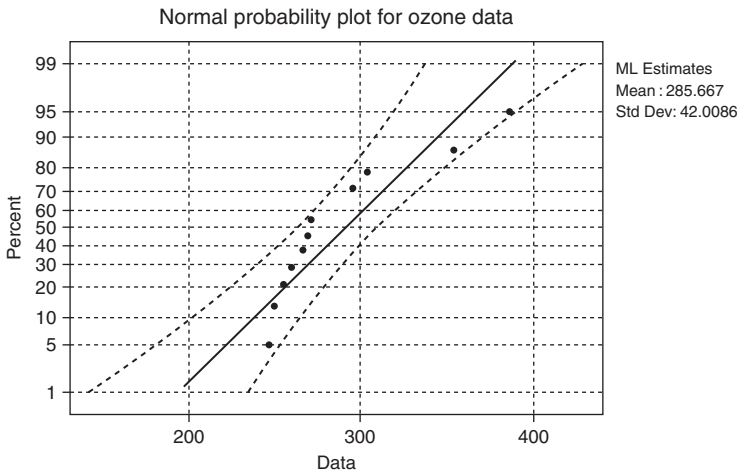
The following data represent the total ozone levels measured in Dobson units at randomly selected locations on the Earth on a particular day:

269	246	388	354	266	303
295	259	274	249	271	254

Can we say that the data are approximately normally distributed? Construct a 95% CI for the population mean μ of ozone levels on this day.

Solution

The following is the probability plot of these data created using Minitab.



Because all the data values lie within the bounds on the normal probability plot (see the discussion in Section 3.2.4), we can assume that the data have approximate normality. We have $\bar{x} = 285.7$ *and* $s = 43.9$. *Also,* $n = 12$. *For* $\alpha = 0.05$, $t_{0.025,\,11} = 2.201$. *A 95% CI for* μ *is:*

$$\bar{x} \pm t_{\alpha/2,(n-1)}\frac{s}{\sqrt{n}} = 285.7 \pm 2.201\left(\frac{43.9}{\sqrt{12}}\right).$$

Hence, a 95% CI for μ, *the average ozone level over the Earth, lies in* (257.81, 313.59).

EXERCISES 5.5

5.5.1. A survey indicates that it is important to pay attention to truth in political advertising. Based on a survey of 1200 people, 35% indicated that they found political advertisements to be untrue; 60% say that they will not vote for candidates whose advertisements are judged to be untrue; and of this latter group, only 15% ever complained to the media or to the candidate about their dissatisfaction.

(a) Find a 95% CI for the percentage of people who find political advertising to be untrue.

(b) Find a 95% CI for the percentage of voters who will not vote for candidates whose advertisements are considered to be untrue.

(c) Find a 95% CI for the percentage of those who avoid voting for candidates whose advertisements are considered untrue and who have complained to the media or to the candidate about the falsehoods in commercials.

(d) For each case above, interpret the results and state any assumptions you have made.

5.5.2. Many mutual funds use an investment approach involving owning stocks whose price/earnings multiples (P/Es) are less than the P/E of the S&P 500. The following data give P/Es of 49 companies that a randomly selected mutual fund owns in a particular year.

6.8	5.6	8.5	8.5	8.4	7.5	9.3	9.4	7.8	7.1
9.9	9.6	9.0	9.4	13.7	16.6	9.1	10.1	10.6	11.1
8.9	11.7	12.8	11.5	12.0	10.6	11.1	6.4	12.3	12.3
11.4	9.9	14.3	11.5	11.8	13.3	12.8	13.7	13.9	12.9
14.2	14.0	15.5	16.9	18.0	17.9	21.8	18.4	34.3	

Find a 98% CI for the mean P/E multiples. Interpret the result and state any assumptions you have made.

5.5.3. Let $X_1, \ldots, X_n$ be a random sample from an $N(\mu, \sigma^2)$ distribution, with σ^2 known.

(a) Show that $\widehat{\mu} = \overline{X}$ is an MLE of the population mean μ.

(b) Show that:

$$P\left(\overline{X} - \frac{2\sigma}{\sqrt{n}} < \mu < \overline{X} + \frac{2\sigma}{\sqrt{n}}\right) = 0.954.$$

(c) Let

$$P\left(\overline{X} - \frac{k\sigma}{\sqrt{n}} < \mu < \overline{X} + \frac{k\sigma}{\sqrt{n}}\right) = 0.90.$$

Find k.

5.5.4. Let the observed mean of a sample of size 45 be $\bar{x} = 68.51$ from a distribution having variance 110. Find a 95% CI for the true mean μ and interpret the result and state any assumptions you have made.

5.5.5. In a random sample of 50 college seniors, 18 indicated that they were planning to pursue a graduate degree. Find a 98% CI for the true proportion of all college seniors planning to pursue a graduate degree, and interpret the result, and state any assumptions you have made.

5.5.6. DVD players coming off an assembly line are automatically checked to make sure they are not defective. The manufacturer wants an interval estimate of the percentage of DVD players that fail the testing procedure. Compute a 90% CI, based on a random sample of size 105 in which 17 DVD players failed the testing procedure. Also, interpret the result and state any assumptions you have made.

5.5.7. Studies have shown that the risk of developing coronary disease increases with the level of obesity, or accumulation of body fat. A study was conducted on the effect of exercise on losing weight. Fifty men who exercised lost an average of 11.4 lb, with a standard deviation of 4.5 lb. Construct a 95% CI for the mean weight loss through exercise. Interpret the result and state any assumptions you have made.

5.5.8. Basing findings on 60 successful pregnancies involving natural birth, an experimenter found that the mean pregnancy term was 274 days, with a standard deviation of 14 days. Construct a 99% CI for the true mean pregnancy term μ.

5.5.9. Let Y be the binomial random variable with parameter p and $n = 400$. If the observed value of Y is $y = 120$, find a 95% CI for p.

5.5.10. For a health screening in a large company, the diastolic and systolic blood pressures of all the employees were recorded. In a random sample of 150 employees, 12 were found to suffer from hypertension. Find 95% and 98% CIs for the proportion of the employees of this company with hypertension.

5.5.11. In a random sample of 500 items from a large lot of manufactured items, there were 40 defectives.

(a) Find a 90% CI for the true proportion of defectives in the lot.

(b) Is the assumption of normal approximation valid?

(c) Suppose we suspect that another lot has the same proportion of defectives as in the first lot. What should be the sample size if we want to estimate the true proportion within 0.01 with 90% confidence?

5.5.12. Pesticide concentrations in sediment from irrigation areas can provide information required to assess the exposure and fate of these chemicals in freshwater ecosystems and their likely impacts on the marine environment. In a study (Müller, J.F., et al., 2000. Pesticides in sediments from Queensland irrigation channels and drains. Mar. Pollut. Bull. 41 (7−12), 294−301), 103 sediment samples were collected from irrigation channels and drains in 11 agricultural areas of Queensland. In 74 of these samples, they detected DDT with concentration levels up to 840 ng/g dw. Obtain a 95% CI for the proportion of total number of sediments with detectable DDT.

5.5.13. Let $\overline{X}$ be the mean of a random sample of size n from an $N(\mu, 16)$ distribution. Find n such that $p(\overline{X} - 2 < \mu < \overline{X} + 2) = 0.95$.

5.5.14. Let X be a Poisson random variable with parameter λ. A sample of 150 observations from this population has a mean equal to 2.5. Construct a 98% CI for λ.

5.5.15. An opinion poll conducted in March of 1996 by a newspaper (*Tampa Tribune*) among eligible voters with a sample size 425 showed that the president, who was seeking reelection, had 45% support. Give a 95% and a 98% CI for the proportion of support for the president.

5.5.16. A random sample of 100 households located in a large city recorded the number of people living in each household, Y, and the monthly expenditure for food, X. The following summary statistics are given:

$$\sum_{i=1}^{100} Y_i = 340$$

$$\sum_{i=1}^{100} Y_i^2 = 1650$$

$$\sum_{i=1}^{100} X_i = 40,000$$

$$\sum_{i=1}^{100} X_i^2 = 44,000,000$$

(a) Form a 95% CI for the mean number of people living in a household in this city.

(b) Form a 95% CI for the mean monthly food expenses.

(c) For each case just given, interpret the results and state any assumptions you have made.

5.5.17. Let $X_1, \ldots, X_n$ be a random sample from an exponential distribution with parameter θ. A sample of 350 observations from this population has a mean equal to 3.75. Construct a 90% CI for θ.

5.5.18. Suppose a coin is tossed 100 times to estimate $p = P$ (*Heads*). It is observed that heads appeared 60 times. Find a 95% CI for p.

5.5.19. Suppose a population of women at least 35 years of age are pregnant with a fetus affected by Down syndrome. We are interested in testing positive on a noninvasive screening test for fetuses affected by Down syndrome by women at least 35 years of age. In an experiment, suppose 52 of 60 women tested positive. Obtain a 95% CI for the true proportion of women at least 35 years of age who are pregnant with a fetus affected by Down syndrome who will receive positive test results from this procedure.

5.5.20. **(a)** Let $X_1, \ldots, X_n$ be a random sample from a Poisson distribution with parameter λ. Derive a $(1 - \alpha)100\%$ large sample CI for λ.

(b) To date nodes in a phylogenetic tree, the mean path length (MPL) is used in estimating the relative age of a node. The following data represent the MPL for 39 nodes (Britton, T. et al., 2002. Phylogenetic dating with confidence intervals using mean path lengths. Mol. Phylogenet. Evol. 24, 58−65). Assume that the data (given in centimeters) follow a Poisson distribution with parameter λ:

65.2	47.0	38.2	13.5	18.0	25.6	16.3	14.0	23.2	18.8
7.5	13.3	11.0	54.9	22.0	50.1	32.6	26.0	13.0	9.0
7.2	4.7	4.5	41.1	45.8	37.0	8.5	30.5	29.3	13.8
7.7	5.5	24.1	12.5	22.3	19.0	9.5	4.7	3.0	

Obtain a 95% CI for λ and interpret.

5.5.21. A person plans to start an Internet service provider in a large city. The plan requires an estimate of the average number of minutes of Internet use of a household in a week. How many households must be (randomly) sampled to be 95% sure that the sample mean is within 15 minutes of the population mean? Assume that a pilot study estimated the value of $\sigma = 35$ minutes.

5.5.22. The fruit fly *Drosophila melanogaster* normally has a gray color. However, because of a mutation a good portion of them are black. A biologist eager to learn about the effects of mutation wants to collect a random sample to estimate the proportion of black fruit flies of this type within 1% error with 95% confidence.

(a) How many individual flies should the researcher capture if there is no information on the population proportion of black flies?

(b) Suppose the researcher has the initial estimate that 25% of the fruit fly *D. melanogaster* have been affected by this mutation. What is the sample size?

5.5.23. In a pharmacological experiment, 35 lab rats were not given water for 11 hours and were then permitted access to water for 1 hour. The amounts of water consumed (mL/h) are given in the following table:

10.6	13.3	15.5	10.7	9.6	12.1	11.8	10.9	9.9	13.2
9.3	11.7	9.9	13.0	12.3	11.0	13.1	11.0	12.5	13.9
14.1	14.8	15.1	12.8	14.0	7.1	14.1	12.7	9.6	12.5
9.0	12.7	13.6	12.5	12.6					

Obtain a 98% CI for the mean amount of water consumed.

5.5.24. In sociology, a social network is defined as the people you make frequent contact with, say, through Facebook. The personal network size for each adult in a random sample of 3000 adults was calculated. The sample had a mean personal network size of 190 with a known population standard deviation of 25. Find a 95% CI for the mean personal network size of all adults to see if we have a normal amount of friends in our network.

5.5.25. **(a)** How does the t distribution compare with the normal distribution?

(b) How does the difference affect the size of CIs constructed using z (normal approximation) relative to those constructed using the t distribution?

(c) Does sample size make a difference?

(d) What assumptions do we need to make in using the t distribution for the construction of a CI?

5.5.26. Use the t table to determine the values of $t_{\alpha/2}$ that would be used in the construction of a CI for a population mean in each of the following cases:
(a) $\alpha = 0.99$, $n = 20$
(b) $\alpha = 0.95$, $n = 18$
(c) $\alpha = 0.90$, n = 25

5.5.27. Let $X_1, \ldots, X_n$ be a random sample from a normal population. A particular realization resulted in a sample mean of 20 with the sample standard deviation 4. Construct a 95% CI for μ when:
(a) $n = 5$, **(b)** $n = 10$, and **(c)** $n = 25$. What happens to the length of the CI as n changes?

5.5.28. In a large university, the following are the ages of 20 randomly chosen employees:

24	31	28	43	28	56	48	39	52	32
38	49	51	49	62	33	41	58	63	56

Assuming that the data came from a normal population, construct a 95% CI for the population mean μ of the ages of the employees of this university. Interpret your answer.

5.5.29. A random sample of size 26 is drawn from a population having a normal distribution. The sample mean and the sample standard deviation from the data are given, respectively, as $\bar{x} = -2.22$ and $s = 1.67$. Construct a 98% CI for the population mean μ and interpret.

5.5.30. A medication is suspected of causing an elevated heart rate in a certain group of high-risk patients. Twenty patients from the group were given the medication. The changes in heart rates were found to be as follows:

−1	8	5	10	2	12	7	9	1	3
4	6	4	12	11	2	−1	10	2	8

Construct a 98% CI for the mean change in heart rate. Assume that the population has a normal distribution. Interpret your answer.

5.5.31. Ten bearings made by a certain process have a mean diameter of 0.905 cm with a standard deviation of 0.0050 cm. Assuming that the data may be viewed as a random sample from a normal population, construct a 95% CI for the actual average diameter of bearings made by this process and interpret.

5.5.32. Air pollution in large US cities is monitored to see whether it conforms to requirements set by the Environmental Protection Agency. The following data, expressed as an air pollution index, give the air quality of a city for 10 randomly selected days:

57.3	58.1	58.7	66.7	58.6	61.9	59.0	64.4	62.6	64.9

Assuming that the data may be looked upon as a random sample from a normal population, construct a 95% CI for the actual average air pollution index for this city and interpret.

5.5.33. To find the average hemoglobin (Hb) level in children with chronic diarrhea, a random sample of 10 children with chronic diarrhea is selected from a city and their Hb levels (g/dL) are obtained as follows:

12.3	11.4	14.2	15.3	14.8	13.8	11.1	15.1	15.8	13.2

Assuming that the data may be looked upon as a random sample from a normal population, construct a 99% CI for the actual average Hb level in children with chronic diarrhea for this city and interpret. Draw a box plot and normal plot for these data, and comment.

5.5.34. Suppose that you need to estimate the mean number of typographical errors per page in the rough draft of a 400-page book. A careful examination of 10 pages gives an average of six errors per page with a standard deviation of two errors. Assuming that the data may be looked upon as a random sample from a normal population, construct a 99% CI for the actual average number of errors per page in this book and interpret. In this problem, is the normal model appropriate?

5.5.35. Creatine kinase (CK) is found predominantly in muscle and is released into the circulation from muscular lesions. Therefore, serum CK activity has been theoretically expected to be useful as a marker in exercise physiology and sports medicine for the detection of muscle injury and overwork. The following data represent the peak CK

activity (measured in IU/L) after 90 min of exercise in 15 healthy young men (Totsuka, M., et al., Break point of serum creatine kinase release after endurance exercise. http://jap.physiology.org/cgi/content/full/93/4/1280):

1112	722	689	251	196	185	128	102	166	178
775	694	514	244	208					

Construct a 95% CI for the mean peak CK activity.

5.5.36. A random sample of 20 observations gave the following summary statistics: $\sum x_i = 234$ and $\sum x_i^2 = 3048$. Assuming that the data may be looked upon as a random sample from a normal population, construct a 95% CI for the actual average, μ.

5.5.37. Let a random sample of size 17 from a normal population for which both mean μ and variance σ^2 are unknown yield $\bar{x} = 3.12$ and $s^2 = 1.04$. Determine a 99% CI for μ.

5.5.38. A random sample from a normal population yields the following 25 values:

90	87	121	96	106	107	89	107	83	92
117	93	98	120	97	109	78	87	99	79
104	85	91	107	89					

(a) Calculate an unbiased estimate $\widehat{\theta}$ of the population mean.
(b) Give an approximate 99% CI for the population mean.

5.5.39. The following are random data from a normal population:

3.3	3.3	4.7	2.6	6.4	4.7	1.7	4.5	5.0	3.0

Construct a 98% CI for the population mean μ.

5.5.40. The following data represent the rates (micrometers per hour) at which a razor cut made in the skin of anesthetized newts is closed by new cells:

28	20	21	39	32	23	18	31	14	23
18	22	28	24	33	12	23	21	25	25

(a) Can we say that the data are approximately normally distributed?
(b) Find a 95% CI for population mean rate μ for the new cells to close a razor cut made in the skin of anesthetized newts.
(c) Find a 99% CI for μ.
(d) Is the 95% CI wider or narrower than the 99% CI? Briefly explain why.

5.5.41. For a particular car, when the brake is applied at 62 mph, the following data give stopping distance (in feet) for 10 random trials on a dry surface (source: http://www.nhtsa.dot.gov/cars/testing/brakes/b.pdf):

146.9	148.4	149.4	148.6	150.3
147.5	147.5	149.3	148.4	145.5

(a) Can we say that the data are approximately normally distributed?
(b) Find a 95% CI for population mean stopping distance μ.

5.5.42. A pharmaceutical company tested a new medicine to be marketed for the treatment of a particular type of virus. To obtain an estimate of the mean recovery time, this medicine was tested on 15 volunteer patients, and the recovery time (in days) was recorded. The following data were obtained:

8	17	10	6	34	11	13	6	9	8
19	4	12	17	7					

(a) Obtain a 95% CI estimate of the mean recovery.
(b) What assumptions do we need to make? Test for these assumptions.

5.6 A confidence interval for the population variance

In this section we derive a CI for the population variance σ^2 based on the chi-square distribution (χ^2 distribution). Recall that the χ^2 distribution, like the Student t distribution, is indexed by a parameter called the degrees of freedom. However, the χ^2 distribution is not symmetric and covers positive values only, and hence, it cannot be used to describe a random variable that assumes negative values. Let $X_1, \ldots, X_n$ be normally distributed with mean μ and variance σ^2, with both μ and σ unknown. We know that:

$$\frac{\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2}$$

has a χ^2 distribution with $(n-1)$ degrees of freedom irrespective of σ^2. Hence, it can be used as a pivot. We now find two numbers, χ^2_L and χ^2_U, such that:

$$P\left(\chi^2_L \le \frac{(n-1)S^2}{\sigma^2} \le \chi^2_U\right) = 1-\alpha.$$

The foregoing inequality can be rewritten as:

$$P\left(\frac{(n-1)S^2}{\chi^2_U} \le \sigma^2 \le \frac{(n-1)S^2}{\chi^2_L}\right) = 1-\alpha.$$

Hence, a $(1-\alpha)100\%$ CI for σ^2 is given by $\left((n-1)S^2/\chi^2_U, (n-1)S^2/\chi^2_L\right)$ For convenience, we take the areas to the right of $\chi^2_U = \chi^2_{\alpha/2}$ and to the left of $\chi^2_L = \chi^2_{1-\alpha/2}$ to be both equal to $\alpha/2$; see Fig. 5.6. Using the chi-square table we can find the values of $\chi^2_{\alpha/2}$ and $\chi^2_{1-\alpha/2}$. Then, we have the following result.

Theorem 5.6.1 *If* $\overline{\mathrm{X}}$ *and* S *are the mean and standard deviation of a random sample of size* n *from a normal population, then:*

$$P\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}} \le \sigma^2 \le \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}}\right) = 1-\alpha,$$

where the χ^2 *distribution has* $(\mathrm{n}-1)$ *degrees of freedom.*

That is, we are $(1-\alpha)100\%$ *confident that the population variance* σ^2 *falls in the interval* $\left((\mathrm{n}-1)\mathrm{S}^2/\chi^2_{\alpha/2},\ (\mathrm{n}-1)S^2/\chi^2_{1-\alpha/2}\right)$

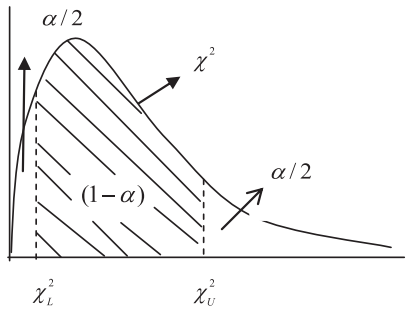


FIGURE 5.6 Chi-square density with equal area on both sides of the confidence interval.

EXAMPLE 5.6.1

A random sample of size 21 from a normal population gave a standard deviation of 9. Determine a 90% CI for σ^2.

Solution

Here $\mathrm{n}=21$ *and* $\mathrm{s}^2=81$. *From the* χ^2 *table with* 20 *degrees of freedom,* $\chi^2_{0.05} = 31.4104$ *and* $\chi^2_{0.95} = 10.8508$. *Therefore, a* 90% *CI for* σ^2 *is obtained from:*